# مسار

برنامج تدريب ميداني

2025

# Introduction to Ethical Hacking Methodologies

# Agenda

- Introduction to Red Team
- Introduction to Penetration Testing
- Penetration Testing Levels
- Penetration Testing vs Vulnerability Assessment
- Penetration Testing Methodology
- Pre-Engagement

# Introduction to Red Team


4

# What is a Red Team?

- **A Red Team** is a group of security professionals who simulate real-world adversaries to assess an organization's ability to detect, respond, and recover from sophisticated attacks.

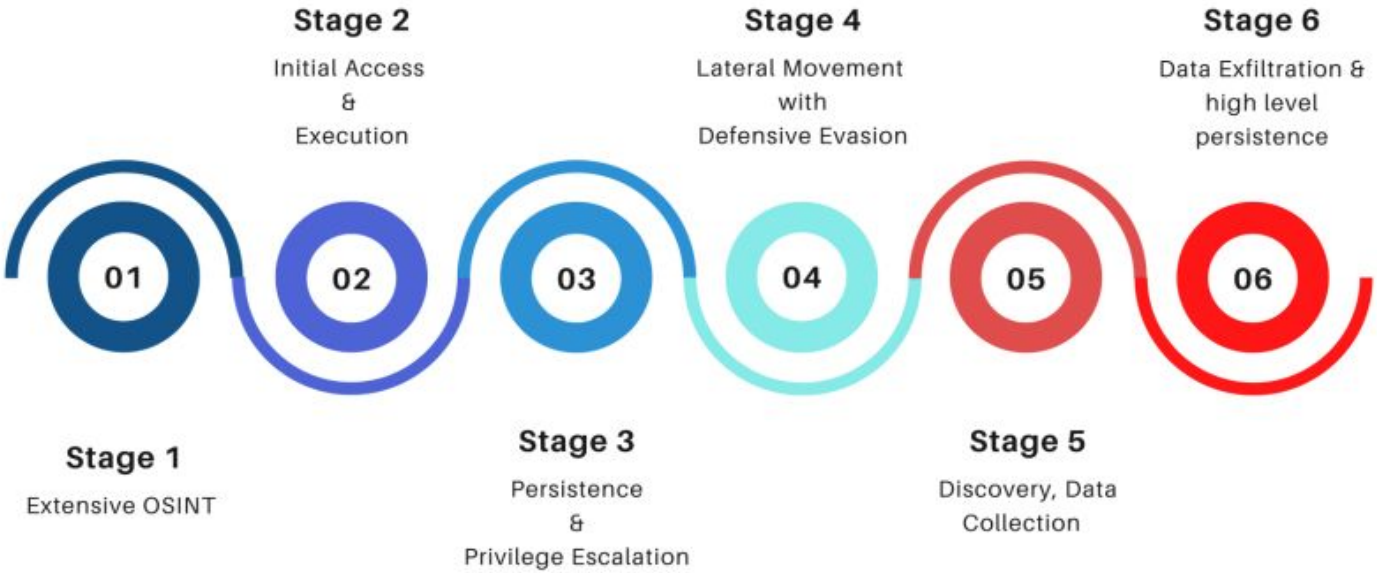
- **Mission**
  - Find vulnerabilities and flaws and demonstrate impact
  - Measure resilience of people, processes, and technology
  - Provide a realistic test of an organization's defenses

- **Key Characteristics**
  - Operates covertly, like real threat actors
  - Focuses on achieving objectives, not just listing flaws and vulnerabilities
  - Uses a mix of technical, physical, and social engineering techniques
  - Often modeled after threat actors (APT groups, insider threats, cybercriminals)

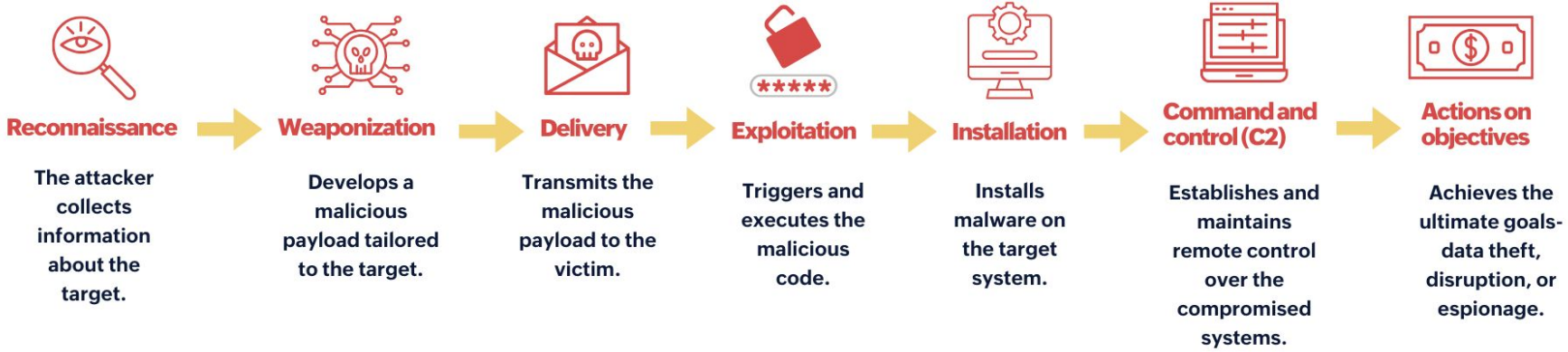
# Red Team vs. Pentesting

| Aspect | Pentesting | Red Teaming |
|---|---|---|
| **Objective** | Identifying Vulnerabilities | Achieving Objectives such as defense evasion |
| **Scope** | Narrow, defined | Broad, goal-based |
| **Timeframe** | Short (1–2 weeks) | Long (months) |
| **Visibility** | Announced | Covert / Black-box |
| **Deliverable** | Vulnerability report | Attack narrative + resilience gaps |

# Red Team Engagement Phases



# The Cyber Kill Chain



# Introduction to Penetration Testing


9

# Introduction to Penetration Testing

- **Penetration testing** is a proactive cybersecurity measure where trained professionals simulate real-world attacks on computer systems, networks, or applications to identify vulnerabilities before they can be exploited by malicious actors
- Penetration testing is also an assessment that is made to determine if the company is secure and can be trusted or need improvement
- Institutes that store critical data such as financial and medical institutes are required to perform an annual or semi-annual penetration test, this will allow the company to always close any security gaps

# Introduction to Penetration Testing

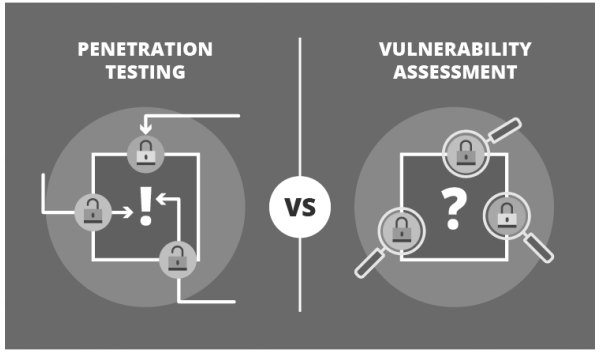
## Objectives of Penetration Testing

- **Confidentiality:** Penetration testing helps ensure that sensitive information remains confidential by identifying weaknesses that could lead to unauthorized access or disclosure of data.
- **Integrity:** Penetration testing assesses the integrity of systems and data by identifying vulnerabilities that could allow for unauthorized modification or tampering.
- **Availability:** Penetration testing helps ensure that systems and services are available when needed by identifying vulnerabilities that could lead to service disruptions, downtime, or denial-of-service attacks.

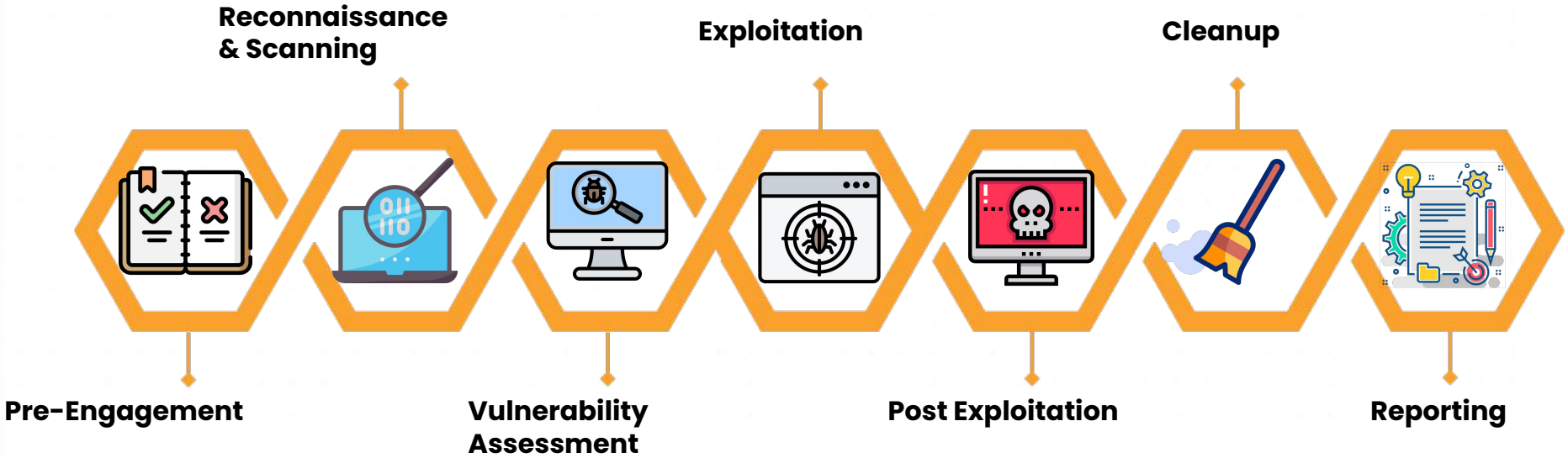
# Penetration Testing Levels

# Penetration Testing vs Vulnerability Assessment

- A **Vulnerability Assessment** is an automated, high-level test that looks for and reports potential vulnerabilities.
- A **Penetration Test** is a detailed hands-on examination by a real person that tries to detect and exploit weaknesses in your system.



# Penetration Testing Methodology



# Pre-Engagement


15

# Pre-Engagement

- **Definition**
  - Initial planning and scoping stage where all objectives, rules of engagement, and expectations are clearly defined between the tester and the client.
- **Pre-engagement Encompasses two Essential Components**
  - Defining the Scope
  - Rules of Engagement

# Pre-Engagement

- **Defining Scope:** This involves clearly defining the objectives, boundaries, and limitations of the penetration test. It identifies the systems, networks, applications, and assets that are within the scope of the test, as well as those that are out of scope.
- Additionally, the scope may specify the types of tests to be conducted, such as external or internal assessments, black-box, grey-box or white-box testing, and specific areas of focus.

# Pre-Engagement

- **Rules of Engagement:** This part establishes the guidelines, procedures, and constraints that govern the conduct of the penetration test.
- It outlines what actions are permitted or prohibited during the testing process, including permissible targets, testing methods, timing, communication protocols, and legal considerations.
- The rules of engagement also address issues related to confidentiality, data protection, and communication with stakeholders to ensure that the test is conducted ethically, safely, and effectively.

# Exercise

The management team at Bazaarjo has decided to conduct a security assessment of their network and cloud infrastructure. However, they are unsure whether to choose a penetration test or a red team engagement. To help them decide, a specialized security vendor provided two sample reports: one from a penetration test and another from a red team exercise.

- Your **task** is to review both reports and write down your observations on the key differences between them.
- After reviewing the reports, create a short briefing (maximum one paragraph) that you would present to Bazaarjo's executive management.
- The briefing should clearly explain, in non-technical language, what each type of assessment offers and why these differences matter for business risk.